Nokia BLR-CORE-02 Major Failure Incident Response - ALARM_2434: Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss)

Node: BLR-CORE-02 | **Vendor:** Nokia | **Severity:** Major | **Alarm Code:** ALARM_2434

Description

This Standard Operating Procedure (SOP) outlines the recommended steps for responding to a Major Failure Incident on the Nokia BLR-CORE-02 (BLR-CORE-02) device, triggered by the ALARM_2434 alarm code. The failure domain is attributed to hardware/environmental faults, which manifest as ASIC memory parity errors, line card crashes, or power supply redundancy loss. The BLR-CORE-02 is a high-capacity, high-performance core node, utilizing Nokia's advanced routing and switching technologies. In this context, the failure is likely caused by an issue with the device's Line Card (LC) or Power Supply Unit (PSU). The Line Card Crash (LCC) scenario may be triggered by a faulty ASIC, which could be due to a hardware defect or excessive temperature fluctuations. The Power Supply Redundancy Loss (PSRL) scenario may be attributed to a faulty PSU or an issue with the Device Redundancy Protocol (DRP). To mitigate these risks, the BLR-CORE-02 is designed with redundant Line Cards and Power Supplies, ensuring uninterrupted service in the event of a failure. However, when such a failure occurs, prompt intervention is necessary to minimize service disruptions and prevent further damage to the device.

Pre-requisite Checks

- Check: Verify the site location of the BLR-CORE-02 (Unknown Site) and ensure that remote hands are dispatched accordingly., Description: Confirm that the site location is accurate and that the necessary personnel have been notified to assist with the incident response.

- Check: Verify the redundant link status (e.g., POS1, POS2) and confirm that they are operational., Description: Ensure that the redundant links are functioning correctly to prevent service disruptions.

Potential Root Causes

Cause	Probability
Faulty Line Card (LC) - ASIC memory parity error	High
Line Card Crash (LCC) due to excessive temperature fluctuations	Medium
Power Supply Redundancy Loss (PSRL) due to faulty PSU or DRP issue	Medium

Diagnostic Commands & Expected Outputs

```
Command: show version | include "ASIC", Expected Output: BLR-CORE-02# show version | include "ASIC", Description: This command isolates the
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ASIC information, including the revision level and any potential
issues with the memory parity checks.

Command: show system redundancy, Expected Output: BLR-CORE-02# show
system redundancy, Description: This command verifies the status of
the device's redundancy features, including the Line Card and Power
Supply redundancy.

Command: show environment temperature, Expected Output: BLR-CORE-02#
show environment temperature, Description: This command checks the
ambient temperature of the device and its immediate environment.
```

Resolution Steps

Step 1: Step: Replace faulty Line Card (LC), Command: replace slot 1:LC-1, Description: Remove the faulty Line Card and replace it with a new one to resolve the ASIC memory parity error.

Step 2: Step: Reset Line Card (LC), Command: reset slot 1:LC-1, Description: Reset the Line Card to its default state to resolve the Line Card Crash (LCC) issue.

Step 3: Step: Replace faulty Power Supply Unit (PSU), Command: replace slot 1:PSU-1, Description: Remove the faulty Power Supply Unit and replace it with a new one to resolve the Power Supply Redundancy Loss (PSRL) issue.

Post-Check Verification

```
Command: show system redundancy, Expected Output: System redundancy is
enabled., Description: Verify that the system redundancy is
functioning correctly after the resolution steps.

Command: show environment temperature, Expected Output: Temperature
within normal operating range., Description: Verify that the ambient
temperature of the device and its immediate environment is within the
normal operating range.
```

Roll-back Procedures

- Step: Restore original configuration, Command: copy /run/config /config/config.txt, Description: Restore the original configuration to the device to ensure that all settings are reverted to their previous state.

- Step: Revert to previous system image, Command: boot image /config/bootimage.txt, Description: Revert the device to its previous system image to ensure that all software is restored to its previous state.